

Project plan

CANASSIST

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March 2026

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1. Introduction

1.1. Background

CanAssist is an organization based in Victoria, British Columbia, Canada, and has been operating for more than 25 years. The organization is dedicated to addressing unmet needs and advancing inclusion, accessibility, and equity for people, particularly individuals with disabilities, on local, provincial, and national levels. CanAssist develops practical technologies and support solutions that improve independence, accessibility, and quality of life for its users. The organization works closely with students, developers, researchers, and healthcare-related professionals to create innovative solutions across areas such as employment, education, and health and wellbeing. Through projects such as CanPlan, CanAssist focuses on creating accessible and user-friendly systems that support people in their daily activities while promoting independence and equal participation in society.

CanPlan is a mobile application developed by CanAssist that helps people with cognitive challenges complete tasks independently. The application allows users to create step-by-step plans using visual and auditory supports such as images, text, audio recordings, and videos. The current version of CanPlan works completely offline and stores all user data locally on the device. While this approach ensures strong privacy and independence from internet connectivity, it also limits certain features such as remote task management, progress tracking, and collaboration with caregivers or support workers.

Additionally, the application interface and branding could benefit from modernization, and accessibility improvements could make the application easier to use for its target audience. This project focuses on researching potential improvements to CanPlan, including usability, accessibility, branding, and possible system expansions. The goal is to explore how the application can better support its users while maintaining privacy, reliability, and accessibility.

1.2. Project Purpose

The purpose of this project is to analyze, redesign, and improve the CanPlan application in order to better support users with cognitive challenges.

The project will focus on:

- Researching the current strengths and limitations of the application
- Improving branding and visual design
- Evaluating accessibility improvements
- Collecting feedback from users through surveys
- Proposing and making new features that could improve the system
- Exploring future system architecture options including possible cloud integration

The project will gradually evolve from research and analysis towards potential implementation improvements.

1.3. Project Scope

In Scope

- Research on current system usage
- Competitor analysis
- Accessibility improvements
- Rebranding and visual improvements
- User feedback through surveys
- Feature proposals and design improvements
- Evaluation of possible future architecture solutions
- Initial exploration of cloud-based features
- Launch an interim release of the application with updated CanAssist branding while maintaining the current functionality
- Development of a prototype cloud backend for internal testing
- Community engagement through requirement gathering and testings

Out of Scope

- Complete redevelopment of the application
- Large scale enterprise deployment
- Integration with external enterprise platforms
- Full production cloud migration without supervisor approval

2. Objectives

The main objectives of this project are:

1. Analyze the current CanPlan application and identify improvement opportunities
2. Conduct research on similar assistive applications and competing solutions
3. Collect feedback from users and stakeholders through surveys, requirement gathering, usability testing, and in-person sessions
4. Improve accessibility and usability of the application while aiming to meet recognized accessibility standards
5. Propose new features that improve user independence
6. Evaluate potential system architecture improvements
7. Prepare the system for potential future cloud integration
8. Deliver documentation and recommendations for future development

3. Project Organisation

3.1. Project Team

Name	Role
Paul Green	Project Supervisor
Wout Jansegers	Software Developer / Student
Brent Pulmans	Academic Mentor
Nick Leonard	App Design Consultant

3.2. Roles and Responsibilities

Project Supervisor

- Provide guidance throughout the internship
- Validate project direction and decisions
- Review progress and deliverables

Developer

- Conduct research and analysis
- Propose improvements and design solutions
- Implement prototypes when needed
- Document the project

Mentor

- Provide feedback and academic support
- Evaluate project results

App Design Consultant

- Provide guidance regarding UI/UX and branding
- Support accessibility-focused design decisions
- Review visual consistency and usability
- Help align the application with CanAssist branding

4. Methodology

The project will follow an iterative development and research approach inspired by Agile principles.

The work will be divided into several phases, starting with research and analysis and gradually moving towards design improvements and possible implementation.

The development will progress step-by-step in order to ensure continuous progress throughout the internship. This approach allows the project scope to evolve over time while ensuring that meaningful results are delivered at each stage.

Regular feedback from the project supervisor will help guide decisions about future development directions.

5. System Overview

The CanPlan system is a mobile application designed to guide users through tasks using structured step-by-step instructions.

Users can create tasks and divide them into smaller steps. Each step can contain multiple types of support content such as:

- Images
- Videos
- Written instructions
- Audio recordings
- Text-to-speech

All data is currently stored locally on the device, which allows the application to function fully offline and ensures user privacy.

Future improvements may introduce optional online functionality while maintaining offline usability.

5.1. System Architecture

Current Architecture

- Mobile application
- Local SQLite database
- Media stored on the device
- Possible Future Architecture

5.2. Proposed Future Architecture

- Mobile application
- Web administration dashboard
- Backend API
- Cloud database

This architecture would allow features such as:

- Remote task assignment
- Progress monitoring
- Multi-user management

However, the implementation of cloud services will depend on decisions made together with the project supervisor.

5.3. Technology Stack

Component	Technology
Mobile Application	Flutter
Programming Language	Dart
Local Database	SQLite
Media Storage	Local device storage

Possible future technologies may include:

Component	Possible Technology
Backend API	Node.js or Spring Boot
Database	PostgreSQL or MySQL
Hosting	Canadian VPS or Cloud Infrastructure

These technologies will be evaluated later in the project if cloud functionality becomes part of the implementation phase.

6. Functional Requirements

The system should allow users to:

- Create tasks
- Divide tasks into step-by-step instructions
- Add images to task steps
- Add video instructions
- Record audio instructions
- Add written instructions
- Track completed tasks and steps
- Store all data locally on the device
- Access tasks without an internet connection

Possible future features may include:

- User accounts
- Role-based access (admin / user)
- Remote task assignment
- Synchronization between devices
- Online progress tracking

7. Non-Functional Requirements

The system must meet several quality requirements.

- **Performance**
 - The application must run smoothly on both Android and iOS devices.
- **Security**
 - User data must remain protected and private.
- **Usability**
 - The application must be easy to understand and use for individuals with cognitive challenges.
- **Reliability**
 - The system must function without requiring an internet connection.
- **Accessibility**
 - The application should follow accessibility principles to ensure it can be used by a wide range of users.

8. Project Planning

8.1. Project Phases

The project will be executed in three main phases. Each phase builds upon the results of the previous one.

Phase 1 – Research, Accessibility and Rebranding

The first phase focuses on understanding the current system and identifying improvements. During this phase research will be conducted on similar applications, and feedback will be collected from users and stakeholders. Accessibility and usability improvements will also be explored, together with a potential rebranding of the application interface.

Activities in this phase include:

- Research and system analysis
- Competitor research
- User feedback collection through surveys
- Accessibility analysis
- Rebranding exploration and design improvements

This phase provides the foundation for the next development phase.

Phase 2 – Feature Design and Implementation

Once the research and design improvements have been completed, the project will move to the development phase. In this stage new features will be designed and implemented based on the insights gathered during the research phase.

Activities in this phase include:

- Designing new application features
- Improving existing functionality
- Implementing new features in the mobile application
- Testing and evaluating usability improvements
- Designing dashboard components that could support caregivers or administrators in managing asks and monitoring progress
- Developing a high-fidelity UI prototype that applies the CanAssist design language created together with the design consultant

The goal of this phase is to improve the functionality and user experience of the CanPlan application while also exploring interface concepts for future dashboard functionality.

Phase 3 – Cloud Architecture Exploration and Implementation

The final phase of the project focuses on exploring the possibility of introducing a cloud-based system for CanPlan.

At the moment, several possible approaches are being considered by the project supervisor. Because of this, the exact architecture has not yet been finalized.

The specific implementation will depend on decisions made together with the project supervisor. If a cloud approach is selected, the project will focus on designing and implementing a first version of this system.

8.2. Timeline

Phase	Estimated Time
Initial Research and System Analysis	Weeks 1–2
Competitor Research	Weeks 1–2
Accessibility Evaluation and Improvements	Weeks 3–4
Rebranding Exploration and Design	Weeks 3–4
User Feedback and Surveys	Weeks 3–6
Feature Design	Weeks 4–5
Feature Implementation	Weeks 5–7
Cloud Architecture Exploration	Weeks 7–8
Cloud System Development	Weeks 8–12
Testing and Improvements	Weeks 11–12
Documentation and Final Report	Weeks 12–13

9. Deliverables

The project will deliver several outputs across the different phases of the internship.

Research Phase Deliverables

- Research report on the current CanPlan system
- Competitor analysis of similar applications
- Accessibility evaluation
- User feedback results from surveys
- Rebranding proposal for the application interface
- Project plan document

Development Phase Deliverables

- Design documentation for new features
- Implementation of selected new features
- Updated application functionality

Cloud Exploration Deliverables

- Evaluation of possible cloud architectures
- Architecture proposal for a cloud-enabled version of CanPlan
- Prototype implementation of cloud features and a prototype cloud backend for internal testing (if approved)

Final Deliverables

- Final project documentation
- Realisation document
- Reflection document
- Updated application prototype
- Recommendations for future development of CanPlan

10. Communication Plan

Communication during the project will take place through:

- Weekly meetings with the project supervisor to discuss progress and planning
- Bi-weekly progress updates sent to the academic mentor
- Daily communication through Microsoft Teams with the project supervisor
- Email communication for formal feedback, document reviews, and scheduling

11. Quality Assurance

Quality will be ensured through:

- Regular supervisor feedback
- User feedback surveys
- Usability evaluation
- Design reviews
- Documentation validation

12. Conclusion

This project aims to analyze and improve the CanPlan application by researching opportunities in usability, accessibility, branding, and functionality. Through research, user feedback, and design improvements, the project will provide recommendations that may guide future development of the application.

A step-by-step approach will ensure continuous progress throughout the internship while allowing flexibility in deciding future development directions such as potential cloud integration.